

Do LLM fingerprints survive outside English?

Interpretable attribution of five frontier models across seven languages, under translation, and against the models themselves as judges

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59%–72%

five-way attribution accuracy from 21 interpretable features, in every one of 7 languages (chance 20%)

p = 0.36

no resource gradient: accuracy does not fall from English to Hindi (cell-level GLM, n = 839)

42/42

cross-lingual transfers above chance: the fingerprint is largely language-invariant (mean 0.49)

ρ = -0.96

models converge stylistically as resources fall: separation shrinks monotonically to Hindi

62%

attribution after Google Translate; English-trained classifier reads translations at 66%

25% vs 20%

only Claude beats chance as a judge of authorship; the other four default to one fixed answer

Why

Provenance tools are built and tested in English, but the need to attribute text to a specific model — disinformation tracing, academic integrity, moderation — is mostly non-English. Detection is known to weaken off English; attribution to a *specific* model has been English-only and black-box.

Nobody had measured attribution with **interpretable** features across a resource gradient, separated model style from language style, tested convergence in low-resource languages, or checked whether translation launders the signal.

Design

5 models: GPT-5.5 · Gemini 3.5 Flash · Claude Opus 4.7 · Grok 4.3 · DeepSeek V4 Pro (one OpenRouter key; served model strings logged).

7 languages in decreasing training-data share: English → Spanish → Chinese → Russian → Japanese → Turkish → Hindi.

12 prompt schemas, each fixing topic, stance (6 for / 6 against) and three sub-claims. A non-subject model (Llama 4) writes a native prompt per language — never a translation — and a third-family reviewer (Qwen) verifies every prompt against its schema. Content is held fixed; only style is free.

840 responses (× 2 generations), no system prompt, temperature 0.7, reasoning traces excluded. 839 pass language ID, length and refusal checks.

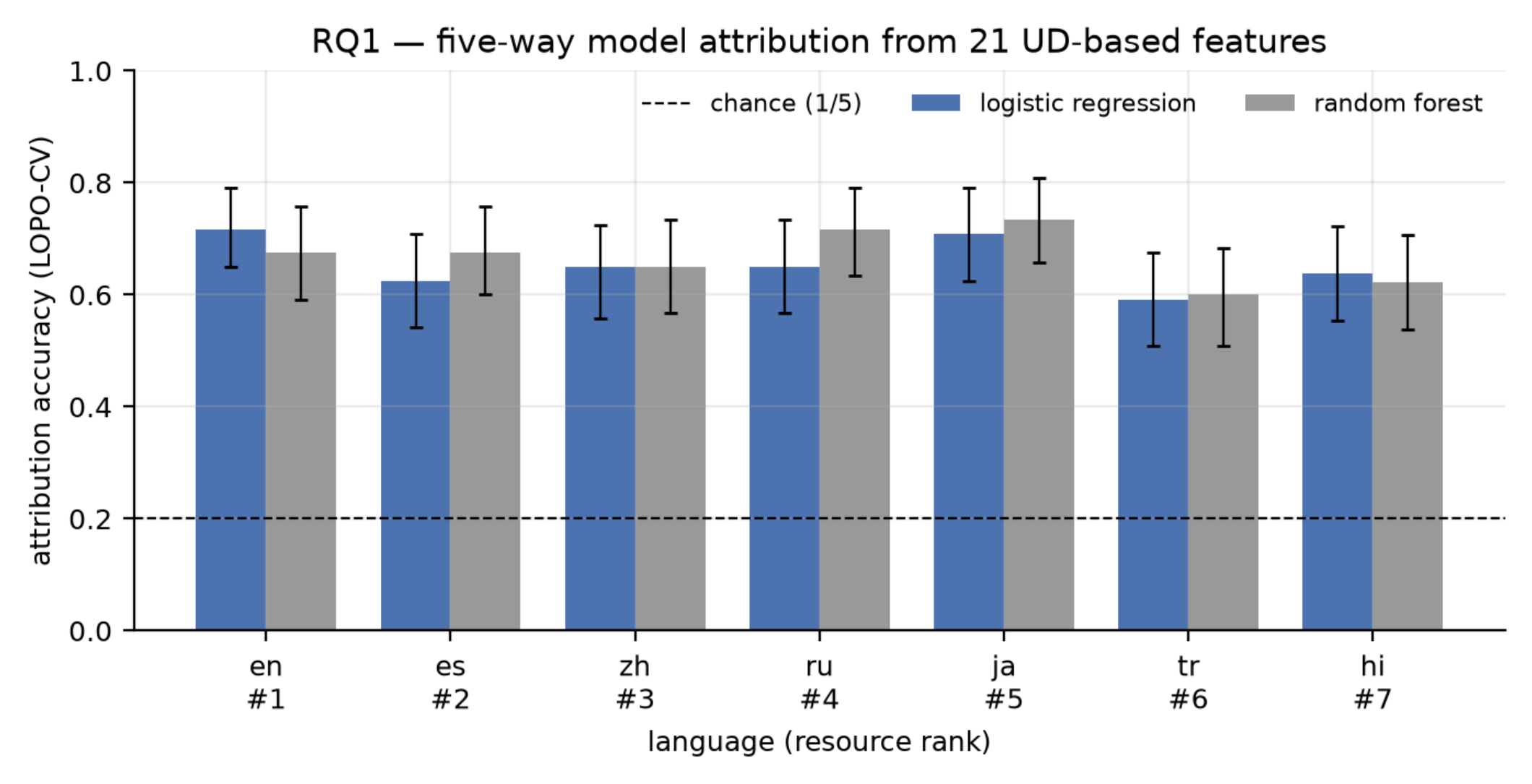
21 features, one definition in every language

All text parsed with Stanza on Universal Dependencies, so each feature means the same thing in Chinese, Turkish or Hindi.

Lexical MATTR · hapax rate · token length · Zipf slope
Syntactic sentence length μ , σ · burstiness · dependency depth · subordination · function-word ratio · first person
Structure paragraph count, length · question rate · connectives
Punctuation comma · colon · dash · semicolon per 1k, script equivalents mapped
Character bigram entropy · digit rate

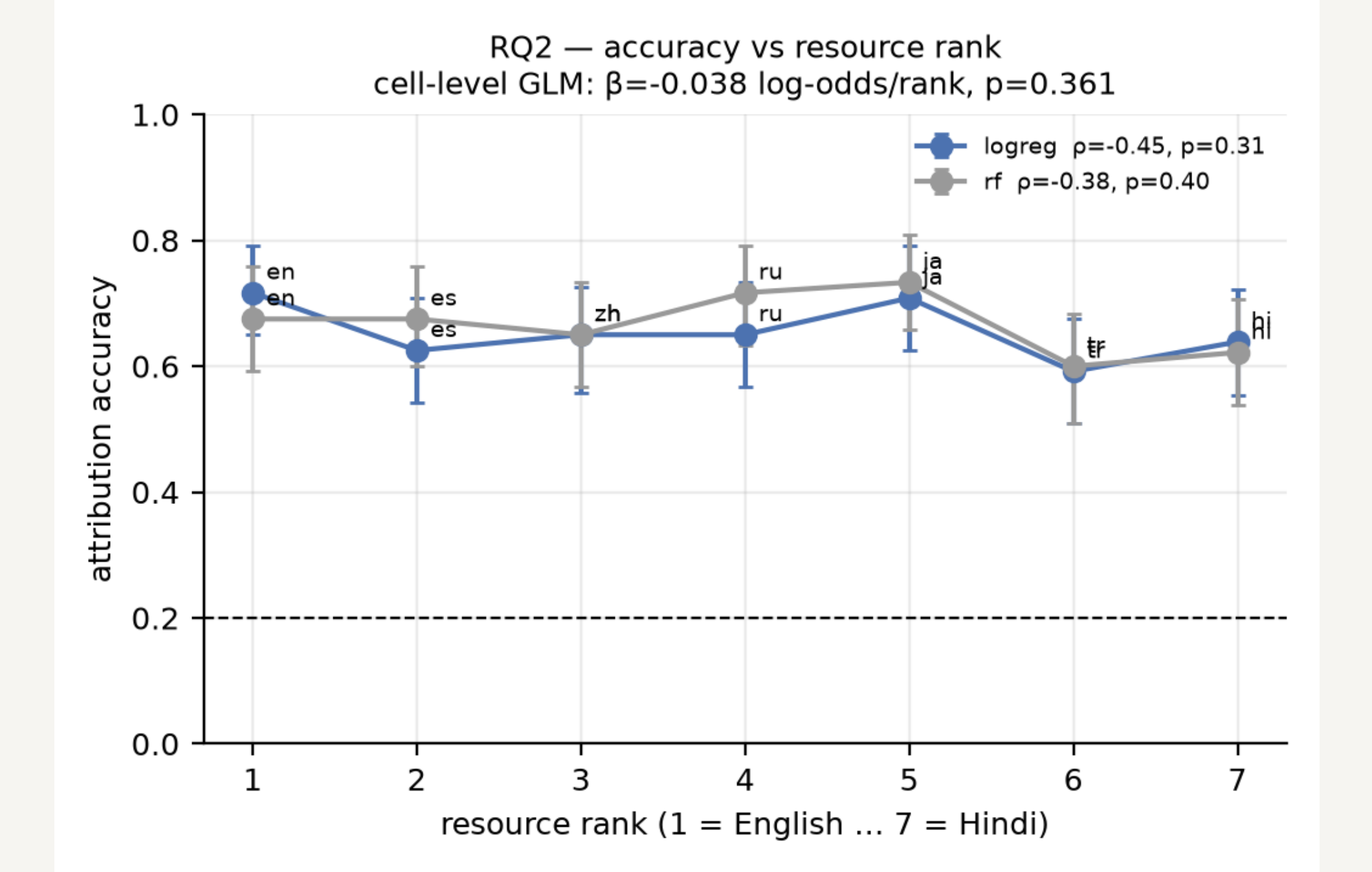
English-only measures (Flesch, Fog, hedges, passive, contractions) are deliberately excluded. All tests pre-specified before collection.

1 · Attribution works in every language



Leave-one-prompt-out CV; logistic regression on standardised features (interpretable) and random forest (ceiling). Every CI excludes chance. GPT and Grok are most identifiable; DeepSeek the blur.

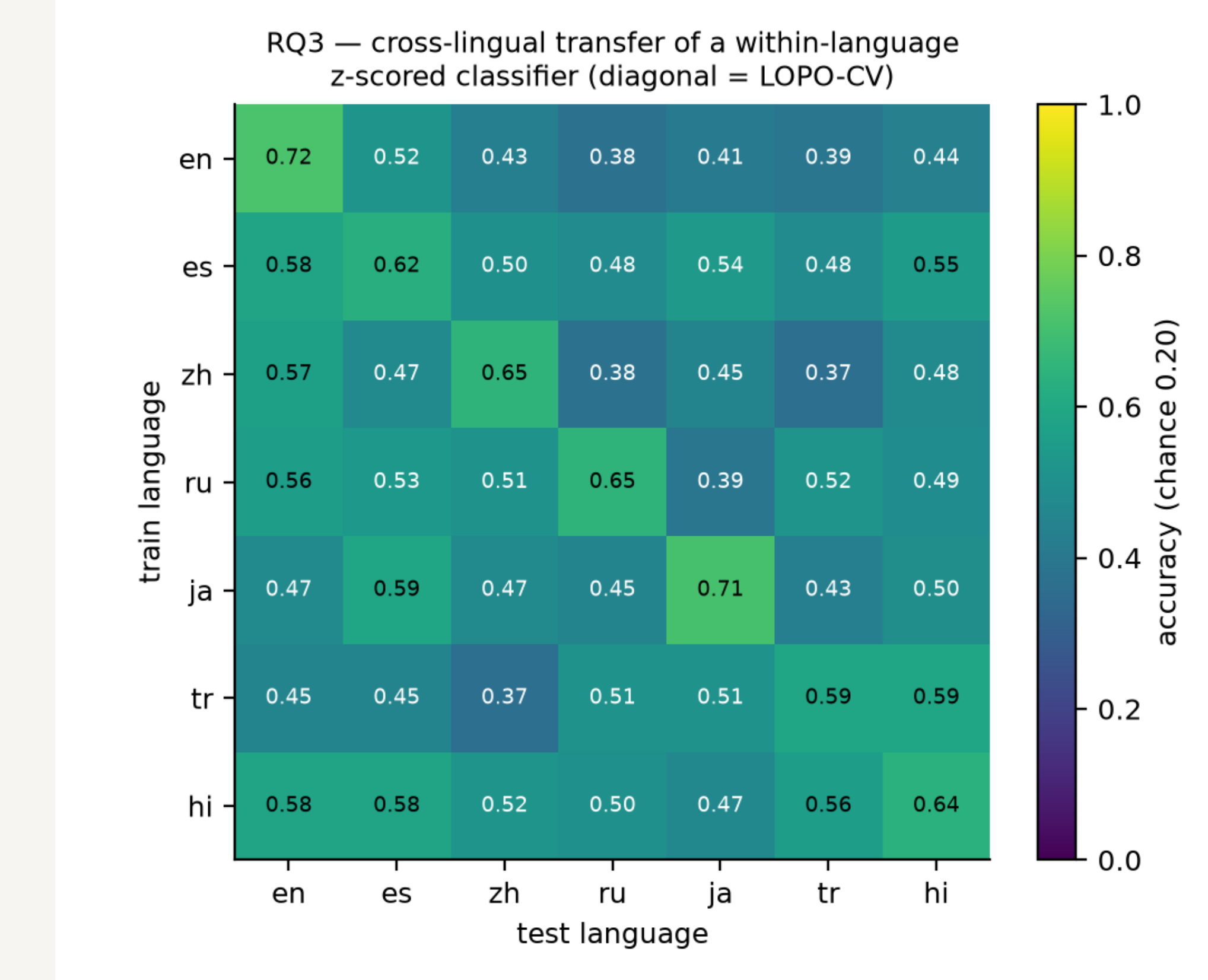
2 · ... and does not fade with resource level



Spearman $\rho = -0.45$ ($p = 0.31$); cell-level GLM $\beta = -0.038$ log-odds per rank ($p = 0.36$). Japanese matches English; Hindi beats Spanish. The pre-registered gradient hypothesis is **not** supported.

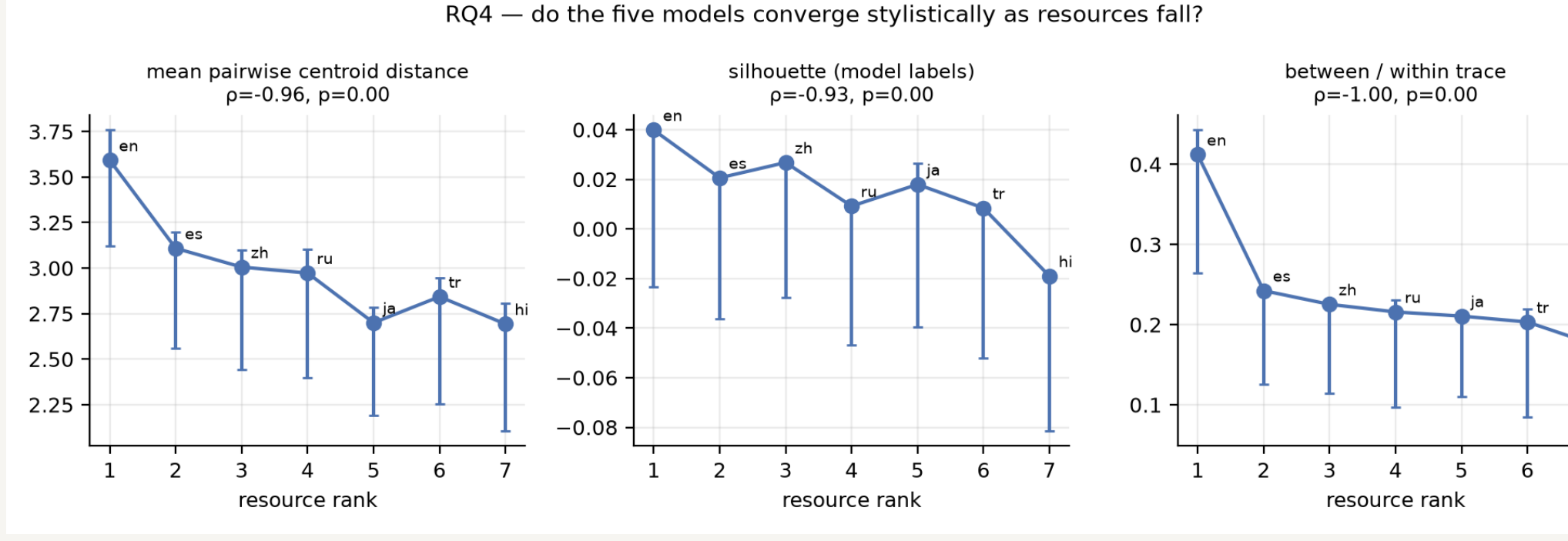
The fingerprint is structural: paragraph count, comma and colon density, and lexical-diversity slope carry it; dependency depth and subordination barely contribute.

3 · Model style is mostly language-invariant



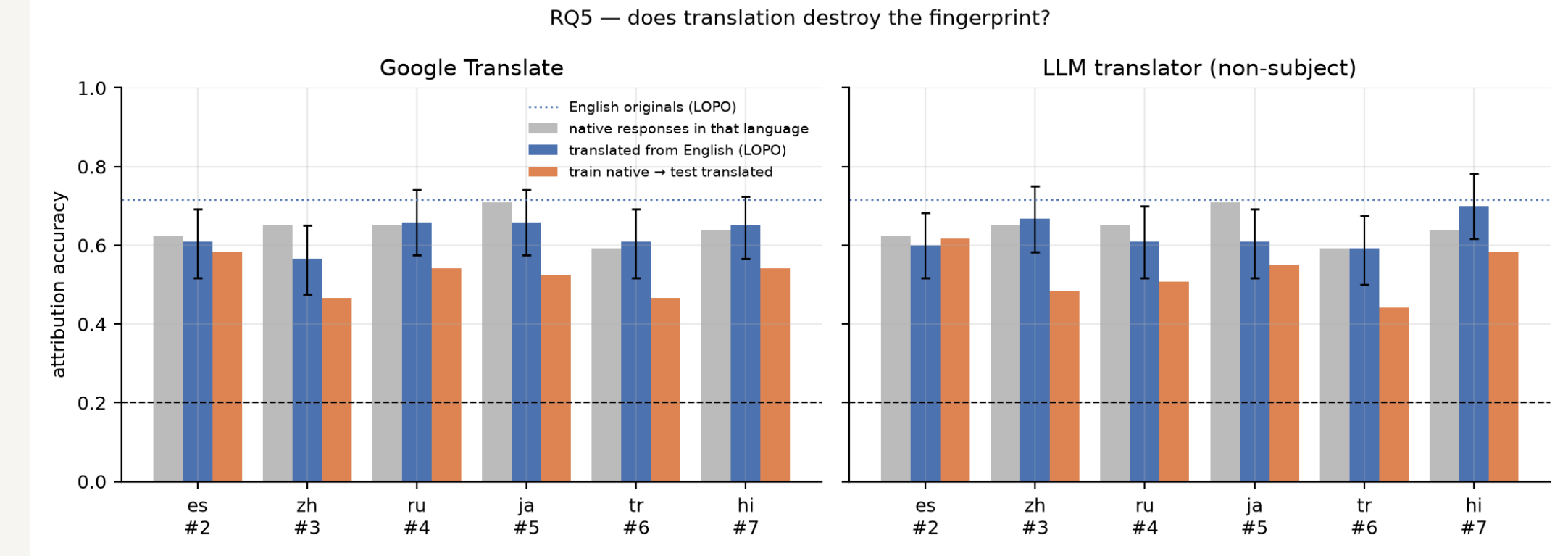
Language explains most feature variance (mean partial η^2 0.54 vs 0.14 for model, 0.13 interaction). But remove each language's mean and a classifier trained on any language beats chance on every other: 42/42 pairs, mean 0.49 vs 0.65 within-language. About three quarters of the fingerprint transfers.

4 · Yet models converge as resources fall



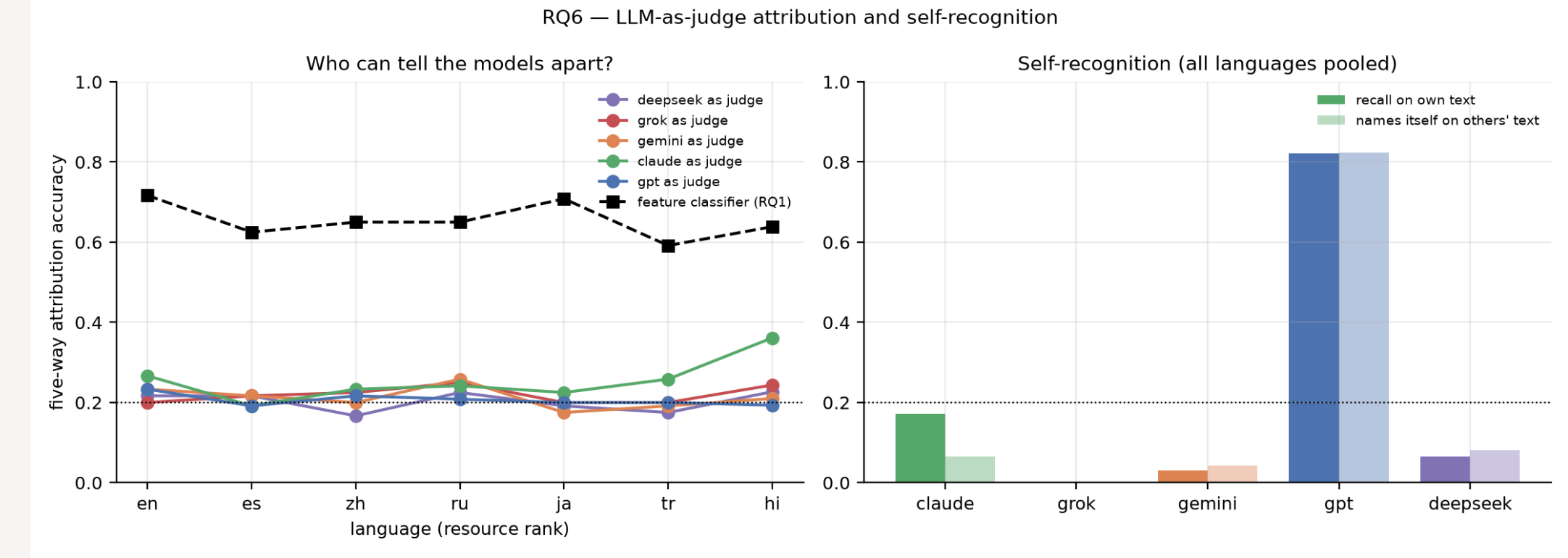
Within-language separation of the five model centroids falls monotonically from English to Hindi (centroid distance $\rho = -0.96$; between/within ratio 0.41 → 0.18). Almost all of it is Grok's terse English register disappearing into the pack. Convergence in the bulk of features, attribution sustained by a few.

5 · Translation does not launder the fingerprint



All 120 English responses translated into the six other languages by Google Translate and by a free open model (MiniMax M3). Attribution on translated text alone: 0.62 / 0.63. An **English-trained** classifier reads the translations at 0.66: structure (paragraph count $\rho = 1.00$, sentence length 0.89) survives; vocabulary is rewritten (MATTR 0.3).

6 · The models themselves cannot do this



Each model asked which of the five wrote each text (reasoning off, options shuffled). GPT 20.6%, Grok 21.9%, Gemini 21.2%, DeepSeek 20.3%: chance, each defaulting to one answer (GPT names itself 82% of the time). Only Claude beats chance (25.4%, $p < 0.001$) with a small self-recognition signal (own-text recall 17% vs 7% false claims), strongest in Hindi.

Takeaways

- Interpretable, UD-based features attribute LLM text in every language and beat every LLM judge by 35–50 points.
- Fingerprints do not fade with resource level and survive translation — provenance across a language boundary works from an English-trained model.
- Low-resource registers are flattening: the five models write more alike in Hindi than in English.
- Limits: one genre; length is part of the linear signal; ordinal resource rank; dated model snapshots.